INCH-POUND
MIL-DTL-27072/25D
w/AMENDMENT 1
23 January 2018
SUPERSEDING
MIL-DTL-27072/25D
8 August 2006

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR
(2 CONDUCTORS) AND SINGLE SHIELDED,
M27072/25JDDA*

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27072.

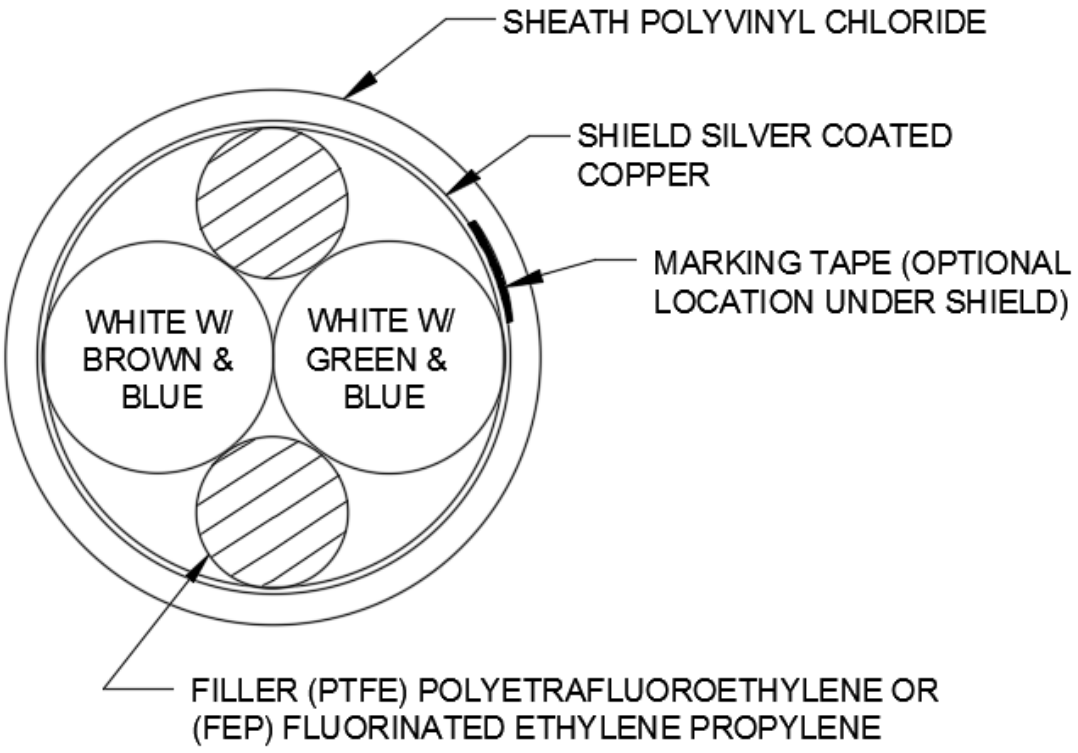


FIGURE 1. Cable configuration.

Note: The Part or Identifying Number (PIN) requires a color code designator (see table I).

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REQUIREMENTS:

Design and construction:	See figure 1
Number of conductors:	2
Temperature rating:	105 °C
Components:	
Basic wire:	SAE AS22759/11-20* (formerly M16878/4BGE***) or NEMA HP-3EWBGE*, (formerly M16878/21BGE***)(see table I)
Conductor:	Silver coated copper
Primary insulation:	Polytetrafluoroethylene
Colors:	White with brown and blue tracers White with green and blue tracers
Jacket over insulation:	Not applicable
Filler:	
Material:	In accordance with MIL-DTL-27072
Lay:	1.0 inch
Cabling:	Not applicable
Binder tape material:	Not applicable
Shield:	
Material:	Silver coated copper
Gauge:	36 AWG
Braid angle:	20 to 40 degrees
Jacket undershield:	Not applicable
Sheath:	
Material:	Polyvinyl chloride
Wall thickness:	.030 inch nominal
Tensile strength:	In accordance with MIL-DTL-27072
Ultimate elongation:	In accordance with MIL-DTL-27072
Accelerated aging:	In accordance with MIL-DTL-27072
Color:	See MIL-DTL-27072
Finished cable:	See table I
Cold bend:	Mandrel size in accordance with MIL-DTL-27072
Dielectric withstanding voltage:	2,000 V rms
Conductor resistance:	In accordance with basic wire specification

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TABLE I. Construction data.

PIN M27072/25	Basic wire except color (reference) SAE AS22579/11-20* or NEMA HP3EWBGE*				Finished cable diameter in. (mm) ±10% 1/
	Wire size (AWG)	Number of strands	Strand size	Nominal diameter inch (mm) 1/	
JDDA*	20	19	32	.058 (1.47)	.197 (5.00)

Asterisks (*) shall be replaced by the appropriate color designator.
1/ Dimensions are in inches. Metric equivalents are provided for information purposes only.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced document. This document references MIL-DTL-27072.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2018-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, please verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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